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Automotive Memory

ADAS, Autonomous and Infotainment - You need memory for everything

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e often only associate memory and storage with our personal computers and smartphones but

with an ever-transforming world that's getting digitized, we see memory requirements skyrocketing in several applications. Automobiles have come a long way from the days when everything was analogue to self-driving cars that have more electronic components than mechanical ones. We speak to Kris Baxter, Corporate Vice President & GM of the Embedded Business Unit at Micron Technology to understand how things have evolved over the years.

How has the memory landscape in automobiles changed over the past 30 years?

Kris: A lot has happened in that time, and it's sort of accelerating as we move into the last, probably 5 or 10 years, but I think one of the things that is most significant is if you go back 30 years, we saw the first introductions of electronics into vehicles were taking place. Now that's hundreds and hundreds of elements of electronics that are making it into vehicles today. The amount of technology, and the timing of which technology is adopted into vehicles, is completely changing. 15 - 20 years ago, it would take five or seven years of running a product into a PC application or mobile application or something before it ever got adopted into



Kris Baxter, Corporate Vice President & GM of the Embedded Business Unit at Micron Technology

a car. It was a pretty rigorous, sort of demonstration of the robustness of the technology before it would ever be adopted into a car. And that's just not what the consumer expects today. That's not what the OEM's are trying to do in regards to the experience that they're trying to provide.

Some of the things that we've seen is that you've gone from what used to be, parts per millions, sort of quality requirement and there was a delay in the timing of introduction of products into cars. Now there's an expectation around a zero-defect methodology, so we're not looking at parts per million but literally how many events are caused. And that drives a significant amount of changes in regards to your methodology or quality management systems. And for our technology to

be adopted sooner into vehicles to deliver the experience that people want from our technology, It means that you have to do something different as a memory supplier to make sure that the technology is ready to go into vehicles and to drive towards a zero-defect methodology much sooner in the ramp of the technology. Also, much sooner in the lifetime of the product, and so that's driven significant changes not only in what we do inside of Micron but also with how we work with our customers, how we collaborate with them, how we understand what they're looking for on the future of their technology roadmaps, how we collaborate into what that future looks like, and making sure that we've got our products and solutions in that space.

So it's pretty dynamic, it's pretty fun to be quite honest because it is such an evolving space and you know it is one of the fastest-growing segments. It's not the largest memory consumer segment in the world, but it is one of the fastest-growing, and so it makes it pretty exciting. Those are just some things that I've seen as far as you know what's happened and what is some of the challenges that we face moving into the future.

You mentioned that there's a lot of difference in the kind of memory that's used in automotive-grade applications compared to what we see in consumer applications, some of them being related to physical parameters like temperature. What are the other parameters required to qualify something as an automotive-grade component?

Kris: So quality is one aspect. For example, if it's in a cell phone and if you have an issue with your phone then you have to reboot it, that's one thing. But if you have that happening in the car, that doesn't work. Also, if you have an issue with your phone

